1.2 Prime and Composite Numbers

Think about the situation:

The teacher gave 5 buttons to Anbuselvan and 6 buttons to Kayalvizhi and asked them to arrange the buttons in all possible rows in such a way that the number of buttons in each row is equal. They did it, in different ways as shown below:

Anbuselvan's Ways	
By arranging 5 buttons in a row, he gets 1 row  1x5=5	By arranging 1 button in each row, he gets 5 rows  5 x1=5
He realises that 5 buttons can be arranged in only 2 rectangular ways. Hence, the only factors of 5 are 1 and 5 (number of rows).	

Kayalvizhi's Ways			
By arranging 6 buttons in a row, she gets 1 row  1x6=6	By arranging 3 buttons in each row, she gets 2 rows  2x3=6	By arranging 2 buttons in each row, she gets 3 rows  3x2 = 6	By arranging 1 button in each row, she gets 6 rows  6 x 1 = 6
She realises that 6 buttons can be arranged in 4 rectangular ways. Hence, the factors of 6 are 1, 2, 3 and 6 (number of rows).			

- Make different rectangular arrangements using 1 button, 2 buttons, 3 buttons, 4 buttons,…, 10 buttons and complete the following table:

Number	Rectangular arrangements possible	Factors	Number of Factors
1		1	
2	1 x 2; 2 x 1	1, 2	2
3			
……			
10	1x 10; 10 x 1; 2x 5; 5 x 2		

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